

Rajesh Phulwaria

Jodhpur, Rajasthan, India — +91-7737346389 — phulwariarajesh152@gmail.com

GitHub — LinkedIn — Portfolio

Profile

AI/ML-focused BCA student interested in understanding intelligent systems from first principles. Experienced in mathematical analysis of neural networks, PyTorch/TensorFlow implementations, Transformer architectures, LLM fundamentals, and Generative AI systems. Published research on gradient flow and neural network optimization, with a focus on translating theoretical understanding into practical AI systems.

Technical Skills

Languages: Python, C++, SQL

Machine Learning: Regression, Classification, Feature Engineering, Preprocessing, EDA, Model Selection, Cross-Validation

Deep Learning: PyTorch, TensorFlow, CNNs, LSTMs, Backpropagation, Gradient Flow, Optimization, Model Evaluation

LLM / Generative AI: Transformers, Self-Attention, Positional Encoding, BPE Tokenization, Embeddings, LLMs, Prompt Engineering, Context Engineering, Function Calling, Hugging Face, Gemini API

Tools / Research: NumPy, Pandas, Git, Linux, Experiment Design, Literature Reading, Reproducible Experiments, Scientific Documentation

“`latex`”

Research & AI Projects

Transformer from Scratch

GitHub — NumPy, PyTorch

- Implemented core Transformer components from first principles, including self-attention, positional encoding, feed-forward networks, masking, and normalization.
- Studied tensor transformations and mathematical operations underlying modern Transformer architectures.

GPT-Style Tokenizer from Scratch

GitHub — Python, BPE

- Implemented a byte-level BPE tokenizer with pair-frequency analysis, iterative merging, vocabulary construction, and encode/decode pipelines.
- Trained on a diverse corpus and analyzed how merge count affects vocabulary size, tokenization granularity, and sequence length.

Research Paper Summarizer

GitHub — Python, LangChain, Groq

- Built an interactive AI-powered research paper summarizer using LangChain, Groq, and GPT-OSS-120B, supporting configurable explanation style and response length.
- Used LangChain's `PromptTemplate` for dynamic prompt construction and Groq for fast LLM inference; designed the project for future extension into a RAG-based paper assistant.

Jarvis – LLM Powered AI Assistant

GitHub — Python, Gemini API

- Built an AI assistant supporting contextual conversations, speech recognition, browser automation, and real-time information retrieval.
- Designed a modular architecture for integrating LLM capabilities and external tools.

Foundational AI Lab

GitHub — PyTorch, TensorFlow

- Developed an open-source repository exploring neural networks from mathematical foundations to practical implementations.
- Implemented optimization techniques, CNNs, LSTMs, learning pipelines, visualization, and experimentation workflows.

Movie Recommendation System

GitHub — Python, TF-IDF, Flask

- Developed a content-based recommendation engine using TF-IDF and cosine similarity with an interactive Flask deployment.

““

Research Publication

A Comparative Analysis of Gradient Flow and Vanishing Gradient Effects in Deep Neural Networks
Zenodo, 2026

- Investigated gradient propagation across deep neural networks and analyzed activation functions affecting optimization stability and learning dynamics.
- DOI: 10.5281/zenodo.19351203

Open Source

fraction-raj (PyPI)

Python

Published an open-source Python package implementing rational arithmetic using Python's data model and operator overloading.

Education

Bachelor of Computer Applications (BCA)

2024–2027

Aishwarya College, Rajasthan

Research Experience

Data Science Intern – Cognifyz Technologies

- Built machine learning pipelines involving preprocessing, feature engineering, experimentation, and model evaluation on real-world datasets.

Python Programming Intern – InternsForge

- Developed Python applications while strengthening software engineering and algorithmic problem-solving skills.